

Kafka Producer Not Sending Messages SOP

Objective

Restore Kafka producer message publishing when an application fails to send events or fails silently.

Scope

Applicable to Kafka-based applications, streaming platforms, backend services, data engineering pipelines, DevOps teams, and platform teams supporting production or synthetic incident workflows.

Pre-requisites

1. Access to Kafka cluster, namespace or broker host.
2. Access to monitoring dashboards such as Grafana, Prometheus, or cloud monitoring.
3. Kafka CLI tools: kafka-topics, kafka-consumer-groups, kafka-console-producer, and kafka-console-consumer.
4. Permission to view application logs, broker logs, and deployment status.
5. Escalation contact for DevOps, platform, and application owner teams.

Incident Metadata

Field	Value
Ticket ID	T034
Issue	Kafka producer not sending messages
Category	Streaming
Service	Kafka
Severity	High
Status	Resolved
Description	Producer fails silently
Expected Resolution	Check broker connectivity and topic existence

Related Log Evidence

No direct matching log entry was present in the provided logs CSV. Use the ticket description, application logs, and monitoring dashboards to validate this incident in runtime.

Procedure

1. Confirm producer failure through application logs, metrics, or missing downstream messages.
2. Validate bootstrap server configuration, topic name, and security settings.
3. Check if Kafka broker is reachable from the producer runtime.
4. Verify topic existence and producer write ACLs.
5. Review producer retries, acks, request timeout, and delivery callback errors.
6. Send a test message using kafka-console-producer from the same network path if possible.
7. Restart or redeploy producer application only after configuration and network checks.
8. Confirm downstream consumers receive new messages and update ticket evidence.

Common Commands

```
kafka-topics.sh --bootstrap-server <broker>:9092 --describe
kafka-consumer-groups.sh --bootstrap-server <broker>:9092 --describe --group <group>
kafka-broker-api-versions.sh --bootstrap-server <broker>:9092
kubectl get pods -n <namespace> | grep kafka
kubectl logs <kafka-pod> -n <namespace> --tail=100
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Producer silent failure	Callback errors not logged	Enable delivery callback logging
Connection timeout	Broker unreachable	Validate network and bootstrap servers
Authorization failed	Missing write ACL	Grant producer ACL
Unknown topic	Topic missing	Create topic or fix topic name

Best Practices

1. Log producer delivery failures explicitly.
2. Use retries with idempotence for critical events.
3. Validate topic and ACLs during deployment.
4. Track producer send rate and error rate.

References

[Apache Kafka Docs](#) | [Internal Logs CSV](#) | [Tickets CSV](#) | [Monitoring Dashboards](#) | [Kafka CLI Runbook](#)